

Some Infinitary Combinatorics

This introductory chapter describes the basic combinatorial tools used in the proofs of most of the results contained in the present monograph.

Our treatment of this classical material is based on two (alternative) principles for regular cardinals: Argyros' ramification lemma (1.1) and the pressing-down lemma (1.3). The main combinatorial results established for regular cardinals—the Erdős–Rado theorem for quasi-disjoint families (1.4) and the Erdős–Rado arrow relations (1.5, 1.7)—are consequences of each of these principles. We need also a simple extension to singular cardinals of the Erdős–Rado theorem for quasi-disjoint families.

We note that sometimes, especially in Chapter 5, we obtain information on calibres of spaces directly from Argyros' ramification lemma (and a technique for singular cardinals due to Tsarpalias), rather than from some derived combinatorial result.

1.1 Lemma. (Argyros' ramification lemma.) Let $\omega \leq \beta \leq \alpha$ with β and α regular, let κ be a cardinal such that $0 < \kappa \ll \beta$, and for every $A \subset \alpha$ with $|A| = \alpha$ let $\mathcal{P}_A$ be a partition of A such that $|\mathcal{P}_A| < \beta$. Then there is a family $\{A_\eta : \eta < \kappa\}$ of subsets of α such that

$$\begin{aligned} |A_\eta| &= \alpha \text{ for } \eta < \kappa, \\ A_{\eta+1} &\in \mathcal{P}_{A_\eta} \text{ for } \eta < \kappa, \\ A_{\eta'} &\subset A_\eta \text{ for } \eta < \eta' < \kappa, \text{ and} \\ \bigcap_{\eta < \kappa} A_\eta &\neq \emptyset. \end{aligned}$$

Proof. We define families $\{\mathcal{A}_\eta : \eta < \kappa\}$ such that

- (i) $\mathcal{A}_0 = \{\alpha\}$;
- (ii) $0 < |\mathcal{A}_\eta| < \beta$ for $\eta < \kappa$;
- (iii) if $A, B \in \mathcal{A}_\eta$ and $A \neq B$, then $A \cap B = \emptyset$ for $\eta < \kappa$;
- (iv) if $A \in \mathcal{A}_\eta$ then $|A| = \alpha$ for $\eta < \kappa$;

- (v) $\mathcal{A}_{\eta+1} \subset \cup \{\mathcal{P}_A : A \in \mathcal{A}_\eta\}$ for $\eta < \kappa$; and
 (vi) $|\alpha \setminus \cup \mathcal{A}_\eta| < \alpha$ for $\eta < \kappa$.

We proceed by recursion.

We define $\mathcal{A}_0$ by (i).

Next for $\eta < \kappa$ we define $\mathcal{A}_{\eta+1}$. We set

$$\begin{aligned}\mathcal{A}'_{\eta+1} &= \cup \{\mathcal{P}_A : A \in \mathcal{A}_\eta\}, \text{ and} \\ \mathcal{A}_{\eta+1} &= \{A \in \mathcal{A}'_{\eta+1} : |A| = \alpha\}.\end{aligned}$$

We verify conditions (ii), (iii), (iv), (v), and (vi) for $\mathcal{A}_{\eta+1}$.

(ii) Let $A \in \mathcal{A}_\eta$. Since $|A| = \alpha$ and $|\mathcal{P}_A| < \beta \leq \alpha$ and $A = \cup \mathcal{P}_A$ and α is regular, there is $B \in \mathcal{P}_A$ such that $|B| = \alpha$; we have $B \in \mathcal{A}_{\eta+1}$ and hence $\mathcal{A}_{\eta+1} \neq \emptyset$. Further, since $|\mathcal{A}_\eta| < \beta$ and $|\mathcal{P}_A| < \beta$ for $A \in \mathcal{A}_\eta$ and β is regular, we have $|\mathcal{A}'_{\eta+1}| < \beta$ and hence $|\mathcal{A}_{\eta+1}| < \beta$.

(iii), (iv) and (v) for $\mathcal{A}_{\eta+1}$ are clear from the definitions.

(vi) For $A \in \mathcal{A}_\eta$ we set $S_A = \cup \{B \in \mathcal{P}_A : |B| < \alpha\}$, and we set $S = \cup \{S_A : A \in \mathcal{A}_\eta\}$. Since α is regular and $|\mathcal{P}_A| < \alpha$ we have $|S_A| < \alpha$ for $A \in \mathcal{A}_\eta$; hence $|S| < \alpha$. Since

$$\alpha \setminus \cup \mathcal{A}_{\eta+1} = (\alpha \setminus \cup \mathcal{A}_\eta) \cup S,$$

we have $|\alpha \setminus \cup \mathcal{A}_{\eta+1}| < \alpha$, as required.

Now we assume that η is a limit ordinal such that $0 < \eta < \kappa$, and that $\mathcal{A}_\xi$ has been defined for $\xi < \eta$, and we define $\mathcal{A}_\eta$. We set

$$\begin{aligned}\mathcal{A}'_\eta &= \left\{ \bigcap_{\xi < \eta} A_\xi : A_\xi \in \mathcal{A}_\xi \text{ and } \bigcap_{\xi < \eta} A_\xi \neq \emptyset \right\}, \text{ and} \\ \mathcal{A}_\eta &= \{A \in \mathcal{A}'_\eta : |A| = \alpha\}.\end{aligned}$$

We verify conditions (ii), (iii), (iv), (v) and (vi) for $\mathcal{A}_\eta$.

(ii) We define $\varphi : \mathcal{A}'_\eta \rightarrow \prod_{\xi < \eta} \mathcal{A}_\xi$ by the rule

$$\varphi\left(\bigcap_{\xi < \eta} A_\xi\right) = \langle A_\xi : \xi < \eta \rangle.$$

It is clear that φ is a one-to-one function. Since $|\mathcal{A}_\xi| < \beta$ for $\xi < \eta$ and $|\eta| < \kappa$ and $\kappa \leq \beta$, we have $|\prod_{\xi < \eta} \mathcal{A}_\xi| < \beta$ by A.5(a); it follows that

$$|\mathcal{A}_\eta| \leq |\mathcal{A}'_\eta| < \beta.$$

We set $S = \cup_{\xi < \eta} (\alpha \setminus \cup \mathcal{A}_\xi)$. Since $|\alpha \setminus \cup \mathcal{A}_\xi| < \alpha$ for $\xi < \eta$ and $|\eta| < \kappa < \alpha$ and α is regular, we have $|S| < \alpha$. Since $\alpha \setminus S = \cup \mathcal{A}'_\eta$ and $|\mathcal{A}'_\eta| < \alpha$, there is $A \in \mathcal{A}'_\eta$ such that $|A| = \alpha$; we have $A \in \mathcal{A}_\eta$ and hence $\mathcal{A}_\eta \neq \emptyset$.

(iii) and (iv) are clear for $\mathcal{A}_\eta$.

(vi) We have

$$\alpha \setminus \bigcup \mathcal{A}_\eta = \bigcup \{A \in \mathcal{A}'_\eta : |A| < \alpha\} \cup S.$$

Since $|\mathcal{A}'_\eta| < \beta \leq \alpha$ and $|S| < \alpha$, we have $|\alpha \setminus \bigcup \mathcal{A}_\eta| < \alpha$, as required.

The definition of the family $\{\mathcal{A}_\eta : \eta < \kappa\}$ is complete.

Since $|\alpha \setminus \bigcup \mathcal{A}_\eta| < \alpha$ for $\eta < \kappa$ and $\kappa < \alpha$ and α is regular, we have $|\bigcup_{\eta < \kappa} (\alpha \setminus \bigcup \mathcal{A}_\eta)| < \alpha$, and hence there is

$$\zeta \in \alpha \setminus \bigcup_{\eta < \kappa} (\alpha \setminus \bigcup \mathcal{A}_\eta).$$

For $\eta < \kappa$ there is $A_\eta \in \mathcal{A}_\eta$ such that $\zeta \in A_\eta$. It is clear that the family $\{A_\eta : \eta < \kappa\}$ satisfies the required conditions.

We apply Lemma 1.1 most frequently with $\beta = \alpha$. In Chapter 5 we will need the case

$$\beta = (\kappa^*)^+ \leq \alpha \text{ with } \kappa \text{ and } \alpha \text{ regular.}$$

The definition of a weakly compact cardinal is given in Appendix A.

1.2 Lemma. Let α be a weakly compact cardinal, and for every $A \subset \alpha$ with $|A| = \alpha$ let $\mathcal{P}_A$ be a partition of A such that $|\mathcal{P}_A| < \alpha$. Then there is a family $\{A_\eta : \eta < \alpha\}$ of subsets of α such that

$$\begin{aligned} |A_\eta| &= \alpha \text{ for } \eta < \alpha, \\ A_{\eta+1} &\in \mathcal{P}_{A_\eta} \text{ for } \eta < \alpha, \text{ and} \\ A_{\eta'} &\subset A_\eta \text{ for } \eta < \eta' < \alpha. \end{aligned}$$

Proof. We define families $\{\mathcal{A}_\eta : \eta < \alpha\}$ such that

- (i) $\mathcal{A}_0 = \{\alpha\}$;
- (ii) $0 < |\mathcal{A}_\eta| < \alpha$ for $\eta < \alpha$;
- (iii) if $A, B \in \mathcal{A}_\eta$ and $A \neq B$, then $A \cap B = \emptyset$ for $\eta < \alpha$;
- (iv) if $A \in \mathcal{A}_\eta$ then $|A| = \alpha$ for $\eta < \alpha$; and
- (v) $\mathcal{A}_{\eta+1} \subset \bigcup \{\mathcal{P}_A : A \in \mathcal{A}_\eta\}$ for $\eta < \alpha$.

(The argument is essentially that of Lemma 1.1. To verify (ii) for limit ordinals η such that $0 < \eta < \alpha$ we set $\gamma = \sup\{|\mathcal{A}_\xi| : \xi < \eta\}$ and we note that since $\gamma < \alpha$ we have

$$|\mathcal{A}_\eta| \leq |\mathcal{A}'_\eta| \leq \prod_{\xi < \eta} |\mathcal{A}_\xi| \leq \gamma^{|\eta|} \leq 2^\gamma \cdot 2^{|\eta|} < \alpha.)$$

Now we set $\mathcal{A} = \bigcup_{\eta < \alpha} \mathcal{A}_\eta$ and we define a partial order $\leq$ on $\mathcal{A}$

by $A \preceq B$ if $A \supset B$. Then $\langle \mathcal{A}, \preceq \rangle$ is a tree of height α with $|\mathcal{A}| < \alpha$ for $\eta < \alpha$; hence there is a branch

$$\Sigma = \{A_\eta : \eta < \alpha\}$$

of $\mathcal{A}$ with $A_\eta \in \mathcal{A}_\eta$ for $\eta < \alpha$. It is clear that the family $\{A_\eta : \eta < \alpha\}$ is as required.

1.3 Lemma. (The pressing-down lemma.) Let $\omega \leq \kappa < \alpha$ with α and κ regular, let

$$S = \{\xi < \alpha : \text{cf}(\xi) \geq \kappa\},$$

and let f be a function from S to α such that $f(\xi) < \xi$ for $\xi \in S$. Then there are $T \subset S$ with $|T| = \alpha$ and $\zeta < \alpha$ such that $f(\xi) < \zeta$ for all $\xi \in T$.

Proof. We suppose the lemma fails. Then for $\zeta < \alpha$ we have $|f^{-1}(\zeta)| < \alpha$ and hence there is $g(\zeta)$ such that

$$\zeta < g(\zeta) < \alpha, \text{ and}$$

$$f(\xi) \geq \zeta \quad \text{for } \xi \in S, \xi \geq g(\zeta).$$

We define $\{\zeta(\eta) : \eta < \kappa\}$ by the rule

$$\zeta(0) = 0,$$

$$\zeta(\eta) = \sup \{\zeta(\eta') : \eta' < \eta\} \text{ for non-zero limit ordinals } \eta < \kappa, \text{ and}$$

$$\zeta(\eta + 1) = g(\zeta(\eta)) \quad \text{for } \eta < \kappa;$$

and we set $\bar{\zeta} = \sup_{\eta < \kappa} \zeta(\eta)$. The function $\eta \rightarrow \zeta(\eta)$ is an ordered-set isomorphism of κ into $\bar{\zeta}$, and since κ is a regular cardinal we have $\text{cf}(\bar{\zeta}) = \kappa$ and hence $\bar{\zeta} \in S$. For $\eta < \kappa$ we have

$$g(\zeta(\eta)) = \zeta(\eta + 1) < \bar{\zeta},$$

and hence $f(\bar{\zeta}) \geq \zeta(\eta)$. From $\bar{\zeta} = \sup_{\eta < \kappa} \zeta(\eta)$ it then follows that $f(\bar{\zeta}) \geq \bar{\zeta}$, a contradiction.

The proof is complete.

We remark, retaining the notation of Lemma 1.3, that since $T = \cup_{\zeta < \bar{\zeta}} (f^{-1}(\{\zeta\}) \cap T)$ and α is regular, there are $T' \subset T$ with $|T'| = \alpha$ and $\zeta < \bar{\zeta}$ such that $f(\xi) = \zeta$ for all $\xi \in T'$.

Definition. An indexed family $\{S_i : i \in I\}$ of sets is a *quasi-disjoint*

family if

$$\bigcap_{i \in I} S_i = S_j \cap S_{j'}, \text{ whenever } j, j' \in I, j \neq j'.$$

It is clear that a family $\{S_i : i \in I\}$ is quasi-disjoint if and only if there is a set S such that

$$S = S_j \cap S_{j'}, \text{ whenever } j, j' \in I, j \neq j'.$$

1.4 Theorem. Let $\omega \leq \kappa \ll \alpha$ with α regular and let $\{S_\xi : \xi < \alpha\}$ be a family of sets such that $|S_\xi| < \kappa$ for $\xi < \alpha$. Then there are $A \subset \alpha$ with $|A| = \alpha$ and a set J such that

$$S_\xi \cap S_{\xi'} = J \text{ for } \xi, \xi' \in A, \xi \neq \xi'.$$

Proof. (In the terminology of the definition above we are to show that there is $A \in [\alpha]^\alpha$ such that $\{S_\xi : \xi \in A\}$ is a quasi-disjoint family. We give two proofs.)

First Proof (using Lemma 1.1). We suppose that if $A \subset \alpha$ and $\{S_\xi : \xi \in A\}$ is quasi-disjoint then $|A| < \alpha$; and for every $A \subset \alpha$ with $|A| = \alpha$ we set

$$J_A = \bigcap_{\xi \in A} S_\xi, \text{ and} \\ \mathcal{B}_A = \{B \subset A : \text{if } \xi, \xi' \in B, \xi \neq \xi', \text{ then } S_\xi \cap S_{\xi'} = J_A\}.$$

Since the set $\mathcal{B}_A$ partially ordered by inclusion is inductive, there is a maximal element $B_A \in \mathcal{B}_A$. We have $|B_A| < \alpha$, and since $\{\xi\} \in \mathcal{B}_A$ for all $\xi \in A$ we have $B_A \neq \emptyset$.

For $\xi \in A \setminus B_A$ it follows from the maximality of B_A that there is $\zeta(\xi) \in B_A$ such that

$$S_\xi \cap S_{\zeta(\xi)} \supsetneq J_A.$$

We define

$$\varphi_A : A \setminus B_A \rightarrow \cup \{\mathcal{P}(S_\zeta) : \zeta \in B_A\}$$

by the rule

$$\varphi_A(\xi) = S_\xi \cap S_{\zeta(\xi)},$$

and we set

$$\mathcal{P}_A = \{B_A\} \cup \{\varphi_A^{-1}(\{S\}) : S \in \cup \{\mathcal{P}(S_\zeta) : \zeta \in B_A\}\}.$$

Then $\mathcal{P}_A$ is a partition of A , and since $|S_\zeta| < \kappa$ for $\zeta \in B_A$ and $\kappa \ll \alpha$, we have $|\mathcal{P}(S_\zeta)| < \alpha$ and hence (since $|B_A| < \alpha$ and α is regular) we

have $|\mathcal{P}_A| < \alpha$. It follows from (the case $\beta = \alpha$ of) Lemma 1.1 that there is a family $\{A_\eta : \eta < \kappa\}$ of subsets of α such that

$$\begin{aligned} |A_\eta| &= \alpha \quad \text{for } \eta < \kappa, \\ A_{\eta+1} &\in \mathcal{P}_{A_\eta} \quad \text{for } \eta < \kappa, \\ A_{\eta'} &\subset A_\eta \quad \text{for } \eta < \eta' < \kappa, \text{ and} \\ \bigcap_{\eta < \kappa} A_\eta &\neq \emptyset. \end{aligned}$$

For $\eta < \kappa$ there is $S(\eta) \in \cup\{\mathcal{P}(S_\xi) : \xi \in B_{A_\eta}\}$ such that

$$\begin{aligned} A_{\eta+1} &= \varphi_{A_\eta}^{-1}(\{S(\eta)\}) \in \mathcal{P}_{A_\eta} \quad \text{and} \\ \varphi_{A_\eta}(\xi) &= S_\xi \cap S_{\xi(\xi)} = S(\eta) \not\supseteq J_{A_\eta} \quad \text{for } \xi \in A_{\eta+1}. \end{aligned}$$

Since $S(\eta) = \varphi_{A_\eta}(\xi) \subsetneq S_\xi$ for $\xi \in A_{\eta+1}$, we have

$$J_{A_\eta} \subsetneq S(\eta) \subset \bigcap_{\xi \in A_{\eta+1}} S_\xi = J_{A_{\eta+1}}$$

and hence $J_{A_{\eta+1}} \setminus J_{A_\eta} \neq \emptyset$ for $\eta < \kappa$; it follows that

$$\left| \bigcup_{\eta < \kappa} J_{A_\eta} \right| \geq \sum_{\eta < \kappa} |J_{A_{\eta+1}} \setminus J_{A_\eta}| \geq \kappa.$$

Now let $\xi \in \bigcap_{\eta < \kappa} A_\eta$. Then $\xi \in A_{\eta+1}$ for $\eta < \kappa$, and from $S_\xi \supset J_{A_\eta}$ we have

$$S_\xi \supset \bigcup_{\eta < \kappa} J_{A_\eta}$$

and hence $|S_\xi| \geq \kappa$, a contradiction.

Second Proof (using Lemma 1.3). We assume without loss of generality that $S_\xi \subset \alpha$ for $\xi < \alpha$. We set

$$S = \{\xi < \alpha : \text{cf}(\xi) \geq \kappa\}$$

and we define $f : S \rightarrow \alpha$ by the rule

$$f(\xi) = \sup(S_\xi \cap \xi) \quad \text{for } \xi \in S.$$

For $\xi \in S$ we have $|S_\xi| < \kappa \leq \text{cf}(\xi)$ and hence $f(\xi) < \xi$. It follows from Lemma 1.3 that there are $T \subset S$ with $|T| = \alpha$ and $\bar{\xi} < \alpha$ such that $f[T] \subset \bar{\xi}$.

For $\xi \in T$ we have $S_\xi \cap \bar{\xi} \in \mathcal{P}_\kappa(\bar{\xi})$, and since $\kappa \ll \alpha$ and α is regular we have $|\mathcal{P}_\kappa(\bar{\xi})| < \alpha$; it follows that there are $T' \subset T$ with $|T'| = \alpha$ and $J \in \mathcal{P}_\kappa(\bar{\xi})$ such that $S_\xi \cap \bar{\xi} = J$ for $\xi \in T'$.

We define a function $\varphi : \alpha \rightarrow T'$ as follows. We set $\varphi(0) = \min T'$,

and if $\xi < \alpha$ and $\varphi(\xi')$ has been defined for all $\xi' < \xi$ we choose $\varphi(\xi) \in T'$ such that

$$\sup \{ \varphi(\xi') : \xi' < \xi \} < \varphi(\xi), \text{ and} \\ \sup \left(\bigcup_{\xi' < \xi} S_{\varphi(\xi')} \right) < \varphi(\xi);$$

such a choice is possible because

$$\left| \bigcup_{\xi' < \xi} S_{\varphi(\xi')} \right| \leq \sum_{\xi' < \xi} |S_{\varphi(\xi')}| \leq |\xi| \cdot \kappa < \alpha$$

and T' is cofinal in α .

We set $A = \{ \varphi(\xi) : \xi < \alpha \}$ and we claim that if $\xi' < \xi < \alpha$ then

$$S_{\varphi(\xi')} \cap S_{\varphi(\xi)} = J.$$

Indeed since $\varphi(\xi'), \varphi(\xi) \in T'$ we have

$$S_{\varphi(\xi')} \cap S_{\varphi(\xi)} \cap \bar{\zeta} = J \cap J = J;$$

and if $\eta \geq \bar{\zeta}$ and $\eta \in S_{\varphi(\xi)}$, then since

$$\sup (S_{\varphi(\xi)} \cap \varphi(\xi)) = f(\varphi(\xi)) < \bar{\zeta},$$

we have $\eta \geq \varphi(\xi) > \sup S_{\varphi(\xi')}$ and hence $\eta \notin S_{\varphi(\xi')}$.

The extension to singular cardinals of the Erdős–Rado theorem for quasi-disjoint families is deferred to Theorem 1.9 below.

Though it is not needed later in this work, we note a strong converse to Theorem 1.4.

Theorem. Let κ and α be cardinals with $\kappa < \alpha$ and $\alpha \geq \omega$. If for every family $\{S_\xi : \xi < \alpha\}$ of sets with $|S_\xi| < \kappa$ for $\xi < \alpha$ there is $A \subset \alpha$ such that $|A| = \alpha$ and $\{S_\xi : \xi \in A\}$ is quasi-disjoint, then $\kappa \ll \alpha$.

Proof. Suppose there are $\beta < \alpha$, $\lambda < \kappa$ such that $\beta^\lambda \geq \alpha$, and let $\{f_\xi : \xi < \alpha\}$ be a subset of β^λ with $f_\xi \neq f_{\xi'}$ for $\xi' < \xi < \alpha$. Then f_ξ is a function from λ to β , and we set

$$S_\xi = \text{graph } f_\xi = \{ \langle \eta, f_\xi(\eta) \rangle : \eta < \lambda \} \quad \text{for } \xi < \alpha.$$

Since $|S_\xi| = \lambda < \kappa$, there is $A \subset \alpha$ such that $|A| = \alpha$ and $\{S_\xi : \xi \in A\}$ is quasi-disjoint. Since $|A| = \alpha > \beta$, for $\eta < \lambda$ the function from A to β defined by $\xi \rightarrow f_\xi(\eta)$ is not one-to-one and hence there are distinct elements ξ, ξ' of A and $\varphi(\eta) < \beta$ such that

$$f_\xi(\eta) = f_{\xi'}(\eta) = \varphi(\eta);$$

it follows that $f_\zeta(\eta) = \varphi(\eta)$ for all $\zeta \in A, \eta < \lambda$ and hence $\{f_\zeta : \zeta \in A\} = \{\varphi\}$. Since $f_\zeta \neq f_{\zeta'}$ for $\zeta, \zeta' \in A$ with $\zeta \neq \zeta'$ we have $|A| = 1$, a contradiction.

Notation. Let α, κ and λ be cardinals and $n < \omega$. The *arrow notation*

$$\alpha \rightarrow (\kappa)_\lambda^n$$

denotes the following partition relation: if

$$[\alpha]^n = \bigcup_{i < \lambda} P_i,$$

then there are $A \subset \alpha$ and $i < \lambda$ such that

$$|A| = \kappa, \text{ and}$$

$$[A]^n \subset P_i.$$

If $A \subset \alpha$ and $[A]^n \subset P_i$, then A is said to be P_i -homogeneous.

1.5 Theorem. Let $\omega \leq \kappa \ll \alpha$ with α regular.

(a) If κ is regular then $\alpha \rightarrow (\kappa)_\lambda^2$ for all $\lambda < \kappa$; and

(b) if κ is singular then $\alpha \rightarrow (\kappa^+)_\kappa^2$.

Proof. (a) Let $[\alpha]^2 = \bigcup_{i < \lambda} P_i$. We are to show that there are $i < \lambda$ and a P_i -homogeneous subset A of α such that $|A| = \kappa$. In fact we show a bit more: for each $i_0 < \lambda$, either there are $i < \lambda$ with $i \neq i_0$ and a P_i -homogeneous subset A of α such that $|A| = \kappa$, or there is a P_{i_0} -homogeneous subset A of α such that $|A| = \alpha$. Without loss of generality we assume in what follows that $i_0 = 0$, and we give two proofs.

First Proof (using Lemma 1.1). We suppose that if $A \subset \alpha$ and A is P_0 -homogeneous then $|A| < \alpha$; and for every $A \subset \alpha$ with $|A| = \alpha$ we set

$$\mathcal{B}_A = \{B \subset A : [B]^2 \subset P_0\}.$$

Since the set $\mathcal{B}_A$ partially ordered by inclusion is inductive, there is a maximal element B_A of $\mathcal{B}_A$. We have $|B_A| < \alpha$, and since $\{\xi\} \in \mathcal{B}_A$ for all $\xi \in A$ we have $B_A \neq \emptyset$.

For $\xi \in A \setminus B_A$ it follows from the maximality of B_A that there is $\zeta(\xi) \in B_A$ such that $\{\xi, \zeta(\xi)\} \notin P_0$. We choose $i(\xi) \in \lambda \setminus \{0\}$ such that

$\{\xi, \zeta(\xi)\} \in P_{i(\xi)}$, and we define

$$\varphi_A : A \setminus B_A \rightarrow B_A \times (\lambda \setminus \{0\})$$

by the rule

$$\varphi_A(\xi) = (\zeta(\xi), i(\xi)),$$

and we set

$$\mathcal{P}_A = \{B_A\} \cup \{\varphi_A^{-1}(\{\zeta, i\}) : \zeta \in B_A, i \in \lambda \setminus \{0\}\}.$$

Since $\mathcal{P}_A$ is a partition of A with $|\mathcal{P}_A| < \alpha$, it follows from (the case $\beta = \alpha$ of) Lemma 1.1 that there is a family $\{A_\eta : \eta < \kappa\}$ of subsets of α such that

$$\begin{aligned} |A_\eta| &= \alpha \quad \text{for } \eta < \kappa, \\ A_{\eta+1} &\in \mathcal{P}_{A_\eta} \quad \text{for } \eta < \kappa, \text{ and} \\ A_{\eta'} &\subset A_\eta \quad \text{for } \eta < \eta' < \kappa. \end{aligned}$$

For $\eta < \kappa$ there are $\bar{\zeta}(\eta) \in B_{A_\eta}$ and $i(\eta) \in \lambda \setminus \{0\}$ such that

$$A_{\eta+1} = \varphi_{A_\eta}^{-1}(\{\bar{\zeta}(\eta), \bar{i}(\eta)\}).$$

Since $\lambda < \kappa$ and κ is regular, there are $I \subset \kappa$ with $|I| = \kappa$ and $\bar{i} \in \lambda \setminus \{0\}$ such that $\bar{i}(\eta) = \bar{i}$ for all $\eta \in I$. We set

$$H = \{\bar{\zeta}(\eta) : \eta \in I\}$$

and we show that

$$\begin{aligned} |H| &= \kappa, \text{ and} \\ [H]^2 &\subset P_{\bar{i}}. \end{aligned}$$

If $\eta, \eta' \in I$ and $\eta < \eta'$, then $\bar{\zeta}(\eta) \in B_{A_\eta}$ and

$$\bar{\zeta}(\eta') \in A_{\eta+1} \subset A_\eta \setminus B_{A_\eta};$$

it follows that $\bar{\zeta}(\eta) \neq \bar{\zeta}(\eta')$. Hence $|H| = |I| = \kappa$. Further, for $\eta, \eta' \in I$ with $\eta < \eta'$ we have

$$\bar{\zeta}(\eta') \in A_{\eta+1} = \varphi_{A_\eta}^{-1}(\{\bar{\zeta}(\eta), \bar{i}(\eta)\})$$

and hence

$$\{\bar{\zeta}(\eta), \bar{\zeta}(\eta')\} \in P_{\bar{i}(\eta)} = P_{\bar{i}};$$

it follows that $[H]^2 \subset P_{\bar{i}}$, as required.

Second Proof (using Lemma 1.3). We set

$$S = \{\xi < \alpha : \text{cf}(\xi) \geq \kappa\}.$$

For $0 < i < \lambda$ and $\xi \in S$ we define a family $\{A_\xi^i(\zeta) : \zeta < \xi\}$ as follows.

$$\begin{aligned} A_\xi^i(0) &= \emptyset & \text{if } \{0, \xi\} \notin P_i \\ &= \{0\} & \text{if } \{0, \xi\} \in P_i; \end{aligned}$$

and if $\zeta < \xi$ and $A_\xi^i(\zeta')$ has been defined for $\zeta' < \zeta$ we set

$$\tilde{A}_\xi^i(\zeta) = \bigcup_{\zeta' < \zeta} A_\xi^i(\zeta')$$

and then

$$\begin{aligned} A_\xi^i(\zeta) &= \tilde{A}_\xi^i(\zeta) & \text{if } \tilde{A}_\xi^i(\zeta) \cup \{\zeta, \xi\} \text{ is not } P_i\text{-homogeneous} \\ &= \tilde{A}_\xi^i(\zeta) \cup \{\zeta\} & \text{if } \tilde{A}_\xi^i(\zeta) \cup \{\zeta, \xi\} \text{ is } P_i\text{-homogeneous.} \end{aligned}$$

We set $A_\xi^i = \bigcup_{\zeta < \xi} A_\xi^i(\zeta)$ and we note that $\xi \notin A_\xi^i$ (since $A_\xi^i \subset \xi$) and $A_\xi^i \cup \{\xi\}$ is P_i -homogeneous.

If there are i and ξ such that $|A_\xi^i| = \kappa$ we set $A = A_\xi^i$ and the proof is complete. We assume in what follows that $|A_\xi^i| < \kappa$ for $0 < i < \lambda$ and $\xi \in S$.

We define $g : S \rightarrow \alpha$ by

$$g(\xi) = \sup\left(\bigcup_{0 < i < \lambda} A_\xi^i\right) \quad \text{for } \xi \in S.$$

Since $\lambda < \kappa \leq \text{cf}(\xi)$ we have $g(\xi) < \xi$ for $\xi \in S$. It follows from Lemma 1.3 that there are $T \subset S$ with $|T| = \alpha$ and $\bar{\zeta} < \alpha$ such that $g[T] \subset \bar{\zeta}$.

For $\xi \in T$ we set $\Phi(\xi) = \langle A_\xi^i : 0 < i < \lambda \rangle$. Writing as usual $\mathcal{P}_\kappa(\bar{\zeta}) = \{B \subset \bar{\zeta} : |B| < \kappa\}$ we have $|\mathcal{P}_\kappa(\bar{\zeta})| < \alpha$ (since $\kappa \leq \alpha$ and α is regular) and hence $|\prod_{0 < i < \lambda} \mathcal{P}_\kappa(\bar{\zeta})| < \alpha$. Since $\Phi(\xi) \in \prod_{0 < i < \lambda} \mathcal{P}_\kappa(\bar{\zeta})$ for $\xi \in T$ there is $T' \subset T$ with $|T'| = \alpha$ such that if $\xi, \xi' \in T'$ then $\Phi(\xi) = \Phi(\xi')$; we show $[T']^2 \subset P_0$.

Let $\xi', \xi \in T'$ with $\xi' < \xi$ and suppose there is i such that $0 < i < \lambda$ and $\{\xi', \xi\} \in P_i$. Since $\Phi(\xi') = \Phi(\xi)$ we have $A_{\xi'}^i = A_\xi^i$; from the fact that the sets

$$A_\xi^i \cup \{\xi\} \text{ and } A_{\xi'}^i \cup \{\xi'\}$$

are P_i -homogeneous it then follows that

$$A_\xi^i \cup \{\xi', \xi\}$$

is P_i -homogeneous. We have

$$\xi' \in A_\xi^i(\xi') \subset A_\xi^i = A_{\xi'}^i,$$

a contradiction.

(b) If κ is singular we have $\kappa^+ \leq \alpha$ from Theorem A.5(c), and

from part (a) (with κ and λ replaced by κ^+ and κ , respectively) we have $\alpha \rightarrow (\kappa^+)_\kappa^2$, as required.

1.6 Corollary. $(2^\alpha)^+ \rightarrow (\alpha^+)_\alpha^2$ for all infinite cardinals α .

Proof. We have $(2^\alpha)^\alpha = 2^\alpha$; thus the (regular) cardinals α^+ , $(2^\alpha)^+$ satisfy the relation $\alpha^+ \ll (2^\alpha)^+$ and the result follows from Theorem 1.5(a).

Notation. Let α be a cardinal and ξ an ordinal. The arrow notation

$$\alpha \rightarrow (\alpha, \xi)^2$$

denotes the following partition relation:

$$[\alpha]^2 = P_0 \cup P_1,$$

then either there is $A \subset \alpha$ with $|A| = \alpha$ such that $[A]^2 \subset P_0$, or there is $A \subset \alpha$ with type $A = \xi$ such that $[A]^2 \subset P_1$.

The negation of the relation $\alpha \rightarrow (\alpha, \xi)^2$ is denoted $\alpha \not\rightarrow (\alpha, \xi)^2$.

1.7 Theorem. If α and κ are cardinals with $\omega \leq \kappa \ll \alpha$ and with α regular, then $\alpha \rightarrow (\alpha, \kappa + 1)^2$.

Proof. Let $[\alpha]^2 = P_0 \cup P_1$. We suppose that if $A \subset \alpha$ and A is P_0 -homogeneous then $|A| < \alpha$; and for every $A \subset \alpha$ with $|A| = \alpha$ we set

$$\mathcal{B}_A = \{B \subset A : [B]^2 \subset P_0\}.$$

Since the set $\mathcal{B}_A$ partially ordered by inclusion is inductive, there is a maximal element B_A of $\mathcal{B}_A$. We have $|B_A| < \alpha$, and since $\{\xi\} \in \mathcal{B}_A$ for all $\xi \in A$ we have $B_A \neq \emptyset$. Since α is regular, there is $\xi(A) < \alpha$ such that $B_A \subset \xi(A)$.

For $\xi \in A \setminus B_A$ it follows from the maximality of B_A that there is $\zeta(\xi) \in B_A$ such that $\{\xi, \zeta(\xi)\} \notin P_0$. We define

$$\varphi_A : A \setminus \xi(A) \rightarrow B_A$$

by the rule

$$\varphi_A(\xi) = \zeta(\xi),$$

and we set

$$\mathcal{P}_A = \{\xi(A) \cap A\} \cup \{\varphi_A^{-1}(\{\zeta\}) : \zeta \in B_A\}.$$

Since $\mathcal{P}_A$ is a partition of A with $|\mathcal{P}_A| < \alpha$, it follows from (the case $\beta = \alpha$ of) Lemma 1.1 that there is a family $\{A_\eta : \eta < \kappa\}$ of subsets

of α such that

$$\begin{aligned} |A_\eta| &= \alpha \text{ for } \eta < \kappa, \\ A_{\eta+1} &\in \mathcal{P}_{A_\eta} \text{ for } \eta < \kappa, \\ A_{\eta'} &\subset A_\eta \text{ for } \eta < \eta' < \kappa, \text{ and} \\ \bigcap_{\eta < \kappa} A_\eta &\neq \emptyset. \end{aligned}$$

For $\eta < \kappa$ there is $\bar{\zeta}(\eta) \in B_{A_\eta}$ such that

$$A_{\eta+1} = \varphi_{A_\eta}^{-1}(\{\bar{\zeta}(\eta)\}).$$

We choose $\bar{\zeta}(\kappa) \in \bigcap_{\eta < \kappa} A_\eta$, we set

$$H = \{\bar{\zeta}(\eta) : \eta \leq \kappa\},$$

and we show that

$$\begin{aligned} \text{type } H &= \kappa + 1, \text{ and} \\ [H]^2 &\subset P_1. \end{aligned}$$

If $\eta < \eta' \leq \kappa$ then $\bar{\zeta}(\eta) \in B_{A_\eta} \subset \xi(A_\eta)$ and

$$\bar{\zeta}(\eta') \in A_{\eta+1} \subset A_\eta \setminus \xi(A_\eta),$$

so that $\bar{\zeta}(\eta) < \bar{\zeta}(\eta')$. Thus $\text{type } H = \kappa + 1$.

For $\eta < \eta' \leq \kappa$ we have

$$\bar{\zeta}(\eta') \in A_{\eta+1} = \varphi_{A_\eta}^{-1}(\{\bar{\zeta}(\eta)\}),$$

so that $\{\bar{\zeta}(\eta'), \bar{\zeta}(\eta)\} \notin P_0$ and hence $\{\bar{\zeta}(\eta), \bar{\zeta}(\eta')\} \in P_1$; it follows $[H]^2 \subset P_1$, as required.

We indicate in Corollary 7.8 below that, at least if the continuum hypothesis is assumed, the statement of Theorem 1.7 cannot be improved for $\alpha = \omega^+$.

1.8 Lemma. Let α be an (infinite) singular cardinal, $\kappa \ll \alpha$ and $\xi < \alpha$ a set of subsets of α such that

$$|S_\xi| < \kappa \text{ for } \xi < \alpha.$$

If $\{\alpha_\sigma : \sigma < \text{cf}(\alpha)\}$ is a set such that

$$\begin{aligned} \alpha_\sigma &\text{ is a regular cardinal for } \sigma < \text{cf}(\alpha), \\ \kappa &\ll \alpha_\sigma \text{ for } \sigma < \text{cf}(\alpha), \\ \text{cf}(\alpha) &< \alpha_{\sigma'} < \alpha_\sigma < \alpha \text{ for } \sigma' < \sigma < \text{cf}(\alpha), \text{ and} \\ \sum_{\sigma < \text{cf}(\alpha)} \alpha_\sigma &= \alpha, \end{aligned}$$

then there are $\{A_\sigma : \sigma < \text{cf}(\alpha)\}$ and $\{J_\sigma : \sigma < \text{cf}(\alpha)\}$ such that

- (i) $A_\sigma \in [\alpha]^{\alpha_\sigma}$ for $\sigma < \text{cf}(\alpha)$,
- (ii) $A_{\sigma'} \cap A_\sigma = \emptyset$ for $\sigma' < \sigma < \text{cf}(\alpha)$, and
- (iii) $S_{\xi'} \cap S_\xi = J_\sigma$ for $\xi', \xi \in A_\sigma, \xi' \neq \xi$,
 $\subset J_\sigma$ for $\xi' \in A_{\sigma'}, \xi \in A_\sigma, \sigma' < \sigma < \text{cf}(\alpha)$.

Proof. There is $\{D_\sigma : \sigma < \text{cf}(\alpha)\} \subset \mathcal{P}(\alpha)$ such that

$$\begin{aligned} |D_\sigma| &= \alpha_\sigma \text{ for } \sigma < \text{cf}(\alpha), \text{ and} \\ D_{\sigma'} \cap D_\sigma &= \emptyset \text{ for } \sigma' < \sigma < \text{cf}(\alpha). \end{aligned}$$

By the Erdős–Rado theorem for quasi-disjoint families there are $C_\sigma \in [D_\sigma]^{\alpha_\sigma}$ and J_σ such that

$$S_{\xi'} \cap S_\xi = J_\sigma \text{ for } \xi', \xi \in C_\sigma, \xi' \neq \xi.$$

We note that if $\sigma' \neq \sigma, \xi' \in C_{\sigma'}$ and $\eta \in S_{\xi'}$, then

$$|\{\xi \in C_\sigma : \eta \in S_\xi, \eta \notin J_\sigma\}| \leq 1.$$

It then follows, setting

$$\begin{aligned} B_\sigma(\sigma') &= \{\xi \in C_\sigma : \text{there is } \xi' \in C_{\sigma'} \text{ such that } S_{\xi'} \cap S_\xi \not\subset J_\sigma\}, \\ B_\sigma &= \cup \{B_\sigma(\sigma') : \sigma' < \sigma\}, \text{ and} \\ A_\sigma &= C_\sigma \setminus B_\sigma, \end{aligned}$$

that

$$\begin{aligned} |B_\sigma(\sigma')| &\leq |\cup \{S_{\xi'} : \xi' \in C_{\sigma'}\}| \leq \alpha_{\sigma'} \cdot \kappa = \alpha_{\sigma'}, \\ |B_\sigma| &\leq \sum_{\sigma' < \sigma} \alpha_{\sigma'} < \alpha_\sigma, \text{ and} \\ |A_\sigma| &= \alpha_\sigma \end{aligned}$$

for $\sigma < \text{cf}(\alpha)$. It is clear that $\{A_\sigma : \sigma < \text{cf}(\alpha)\}$ and $\{J_\sigma : \sigma < \text{cf}(\alpha)\}$ are as required.

Conditions sufficient to ensure that there is a set $\{\alpha_\sigma : \sigma < \text{cf}(\alpha)\}$ as in the statement of Lemma 1.8 are given in Theorem A.7 of the Appendix. One such condition is that $\kappa = \omega$; we use Lemma 1.8 with $\kappa = \omega$ only, in Theorem 8.10 below.

The next result is the extension to singular cardinals of the Erdős–Rado theorem.

1.9 Theorem. Let α be singular with $\text{cf}(\alpha) > \omega$, let $\kappa \ll \alpha$ and

$\kappa \ll \text{cf}(\alpha)$, and let $\{S_\xi : \xi < \alpha\}$ be a set of subsets of α such that

$$|S_\xi| < \kappa \quad \text{for } \xi < \alpha.$$

If $\{\alpha_\sigma : \sigma < \text{cf}(\alpha)\}$ is a set such that

α_σ is a regular cardinal for $\sigma < \text{cf}(\alpha)$,

$\kappa \ll \alpha_\sigma$ for $\sigma < \text{cf}(\alpha)$,

$\text{cf}(\alpha) < \alpha_{\sigma'} < \alpha_\sigma < \alpha$ for $\sigma' < \sigma < \text{cf}(\alpha)$, and

$$\sum_{\sigma < \text{cf}(\alpha)} \alpha_\sigma = \alpha,$$

then there are $\Gamma, J, \{A_\sigma : \sigma \in \Gamma\}$ and $\{J_\sigma : \sigma \in \Gamma\}$ such that

- (i) $\Gamma \in [\text{cf}(\alpha)]^{\text{cf}(\alpha)}$,
- (ii) $A_\sigma \in [\alpha]^{\alpha_\sigma}$ for $\sigma \in \Gamma$,
- (iii) $A_\sigma \cap A_{\sigma'} = \emptyset$ for $\sigma', \sigma \in \Gamma, \sigma' \neq \sigma$, and
- (iv) $S_{\xi'} \cap S_\xi = J_\sigma$ for $\xi', \xi \in A_\sigma, \xi' \neq \xi$
 $= J$ for $\xi' \in A_{\sigma'}, \xi \in A_\sigma, \sigma, \sigma' \in \Gamma, \sigma \neq \sigma'$.

Proof. From Lemma 1.8 there are $\{C_\sigma : \sigma < \text{cf}(\alpha)\}$ and $\{J_\sigma : \sigma < \text{cf}(\alpha)\}$ with properties (analogous to) (ii) and (iii) and such that

$$\begin{aligned} S_{\xi'} \cap S_\xi &= J_\sigma \quad \text{for } \xi', \xi \in C_\sigma, \xi' \neq \xi \\ &\subset J_\sigma \quad \text{for } \xi' \in C_{\sigma'}, \xi \in C_\sigma, \sigma' < \sigma < \text{cf}(\alpha). \end{aligned}$$

Since $\kappa \ll \text{cf}(\alpha)$ and $|J_\sigma| < \kappa$ for $\sigma < \text{cf}(\alpha)$ there are (by the Erdős–Rado theorem for quasi-disjoint families) Γ and J such that

$$\begin{aligned} \Gamma &\in [\text{cf}(\alpha)]^{\text{cf}(\alpha)} \text{ and} \\ J_{\sigma'} \cap J_\sigma &= J \quad \text{for } \sigma', \sigma \in \Gamma, \sigma' \neq \sigma. \end{aligned}$$

We consider two cases.

Case 1. $J = \emptyset$. We note that if $\sigma', \sigma \in \Gamma, \sigma' \neq \sigma$ and $\eta \in J_{\sigma'}$, then

$$|\{\xi \in C_\sigma : \eta \in S_\xi\}| \leq 1$$

(since $\eta \notin J_\sigma \cap J_{\sigma'} = \emptyset$). It then follows, setting

$$\begin{aligned} B_\sigma(\sigma') &= \{\xi \in C_\sigma : S_\xi \cap J_{\sigma'} \neq \emptyset\}, \\ B_\sigma &= \cup \{B_\sigma(\sigma') : \sigma' \in \Gamma, \sigma' \neq \sigma\}, \text{ and} \\ A_\sigma &= C_\sigma \setminus B_\sigma, \end{aligned}$$

that

$$\begin{aligned} |B_\sigma(\sigma')| &\leq |J_{\sigma'}| < \kappa, \\ |B_\sigma| &\leq |\Gamma| \cdot \kappa < \alpha_0, \text{ and} \\ |A_\sigma| &= \alpha_0 \end{aligned}$$

for $\sigma \in \Gamma$. It is clear that

$$S_\xi \cap J_{\sigma'} = \emptyset \quad \text{for } \xi \in A_\sigma, \sigma, \sigma' \in \Gamma, \sigma \neq \sigma'.$$

Now let $\xi' \in A_{\sigma'}$, $\xi \in A_\sigma$, $\sigma' \in \Gamma$, and $\sigma' < \sigma$. Then $S_{\xi'} \cap S_\xi \subset J_{\sigma'}$, and from $S_{\xi'} \cap J_{\sigma'} = \emptyset$ we have

$$S_{\xi'} \cap S_\xi = \emptyset,$$

as required.

Case 2. $J \neq \emptyset$. We set $\tilde{J} = \emptyset$, $\tilde{J}_\sigma = J_\sigma \setminus J$ for $\sigma \in \Gamma$ and $\tilde{S}_\xi = S_\xi \setminus J$ for $\xi \in C_\sigma$, $\sigma \in \Gamma$, and we note from Case 1 that there is $\{A_\sigma : \sigma \in \Gamma\}$ satisfying (ii) and (iii), and

$$\begin{aligned} \tilde{S}_{\xi'} \cap \tilde{S}_\xi &= \tilde{J}_\sigma \quad \text{for } \xi', \xi \in A_\sigma, \xi' \neq \xi \\ &= \tilde{J} \quad \text{for } \xi' \in A_{\sigma'}, \xi \in A_\sigma, \sigma, \sigma' \in \Gamma, \sigma \neq \sigma'. \end{aligned}$$

It is clear that condition (iv) also is satisfied.

The proof is complete.

We note that, according to the remark following A.7 of the Appendix the hypothesis $\kappa \ll \alpha$ in Lemma 1.8 and Theorem 1.9 is redundant; that is, the condition $\kappa \ll \alpha$ follows from the existence of the given family $\{\alpha_\sigma : \sigma < \text{cf}(\alpha)\}$.

1.10 Corollary. Let $\omega \leq \kappa \ll \alpha$ and $\kappa \ll \text{cf}(\alpha)$, let $\{S_\xi : \xi < \alpha\}$ be a set of sets such that

$$|S_\xi| = \lambda < \kappa \text{ for } \xi < \alpha,$$

and for $\xi < \alpha$ let φ_ξ be a one-to-one function from S_ξ onto λ . Then there is $A \subset \alpha$ with $|A| = \alpha$ such that if $\xi, \xi' \in A$ and $\eta \in S_\xi \cap S_{\xi'}$, then $\varphi_\xi(\eta) = \varphi_{\xi'}(\eta)$.

Proof. Case 1. α is regular. From the Erdős–Rado theorem on quasi-disjoint families (Theorem 1.4) we may assume without loss of generality that there is a set J such that $S_\xi \cap S_{\xi'} = J$ for $\xi' < \xi < \alpha$. Then

$$\{\varphi_\xi \upharpoonright J : \xi < \alpha\} \subset \lambda^J,$$

and since $|J| \leq \lambda < \kappa < \alpha$ and $\kappa \ll \alpha$ we have $|\lambda^J| < \alpha$. Since α is regular, there is $A \in [\alpha]^\alpha$ such that $\varphi_\xi|J = \varphi_{\xi'}|J$ for $\xi, \xi' \in A$, as required.

Case 2. α is singular. From Theorem A7 there is a set $\{\alpha_\sigma : \sigma < \text{cf}(\alpha)\}$ of regular cardinals such that

$$\kappa \ll \alpha_\sigma < \alpha \quad \text{for } \sigma < \text{cf}(\alpha) \text{ and}$$

$$\sum_{\sigma < \text{cf}(\alpha)} \alpha_\sigma = \alpha,$$

and by the extension to singular cardinals of the Erdős–Rado theorem there are $\Gamma, J, \{A_\sigma : \sigma \in \Gamma\}$ and $\{J_\sigma : \sigma \in \Gamma\}$ such that

$$\Gamma \in [\text{cf}(\alpha)]^{\text{cf}(\alpha)},$$

$$A_\sigma \in [\alpha_\sigma]^{\alpha_\sigma} \quad \text{for } \sigma \in \Gamma,$$

$$A_\sigma \cap A_{\sigma'} = \emptyset \quad \text{for } \sigma', \sigma \in \Gamma, \sigma' \neq \sigma, \text{ and}$$

$$S_\xi \cap S_{\xi'} = J_\sigma \quad \text{for } \xi, \xi' \in A_\sigma, \xi \neq \xi'$$

$$= J \quad \text{for } \xi \in A_\sigma, \xi' \in A_{\sigma'}, \sigma, \sigma' \in \Gamma, \sigma \neq \sigma'.$$

It follows from Case 1 (with α_σ replacing α) that for $\sigma \in \Gamma$ there is $B_\sigma \in [A_\sigma]^{\alpha_\sigma}$ such that if $\xi, \xi' \in B_\sigma$ and $\eta \in S_\xi \cap S_{\xi'}$, then $\varphi_\xi(\eta) = \varphi_{\xi'}(\eta)$. For $\sigma \in \Gamma$ we define a function ψ_σ from J_σ to λ by the rule

$$\psi_\sigma = \varphi_\xi|J_\sigma \quad \text{for } \xi \in B_\sigma.$$

Since $\psi_\sigma[J_\sigma] \subset \lambda$ and $|\psi_\sigma[J_\sigma]| \leq \lambda < \kappa \ll \text{cf}(\alpha)$, there are $\Delta \in [\Gamma]^{\text{cf}(\alpha)}$ and a set $M \subset \lambda$ such that $\psi_\sigma[J_\sigma] = M$ for $\sigma \in \Delta$. Since ψ_σ is a one-to-one function from J_σ onto M , there is by Case 1 (with $\text{cf}(\alpha)$ replacing α) a set $E \subset \Delta$ with $|E| = \text{cf}(\alpha)$ such that if $\sigma, \sigma' \in E$ and $\eta \in J_\sigma \cap J_{\sigma'}$, then $\psi_\sigma(\eta) = \psi_{\sigma'}(\eta)$. We set $A = \cup_{\sigma \in E} B_\sigma$ and we claim that if $\xi, \xi' \in A$ and $\eta \in S_\xi \cap S_{\xi'}$, then $\varphi_\xi(\eta) = \varphi_{\xi'}(\eta)$. If there is $\sigma \in E$ such that $\xi, \xi' \in B_\sigma$ this is clear; and if $\xi \in B_\sigma, \xi' \in B_{\sigma'}$ with $\sigma, \sigma' \in E$ and $\sigma \neq \sigma'$, then $\eta \in J_\sigma \cap J_{\sigma'} = J$ and hence

$$\varphi_\xi(\eta) = \psi_\sigma(\eta) = \psi_{\sigma'}(\eta) = \varphi_{\xi'}(\eta),$$

as required.

1.11 Theorem. Let $\alpha \geq \omega$ with α regular, ρ an ordinal such that $\rho < \alpha$, and $f: \alpha \rightarrow \mathcal{P}(\alpha)$ such that

$$\text{type } f(\xi) \leq \rho \quad \text{for } \xi < \alpha, \text{ and}$$

$$\xi \notin f(\xi) \quad \text{for } \xi < \alpha.$$

Then there is $A \subset \alpha$ such that

$$|A| = \alpha, \text{ and} \\ \xi \notin f(\xi) \quad \text{for } \xi, \zeta \in A$$

(i.e., A is free for f).

Proof. Suppose there is no set A satisfying the conclusion of the theorem. We define families $\{\zeta_\eta : \eta \leq \rho\}$, $\{A_\eta : \eta \leq \rho\}$, $\{B_\eta : \eta \leq \rho\}$, and $\{\bar{\zeta}_\eta : \eta \leq \rho\}$ recursively as follows. If $\eta \leq \rho$ and if $\zeta_{\eta'}$, $A_{\eta'}$, $B_{\eta'}$, and $\bar{\zeta}_{\eta'}$ have been defined for $\eta' < \eta$, we set

$$\zeta_\eta = \sup_{\eta' < \eta} \bar{\zeta}_{\eta'},$$

we choose for A_η a subset of $\alpha \setminus \zeta_\eta$ which is maximal (among all subsets of $\alpha \setminus \zeta_\eta$) with respect to the property that A_η is free for f , we set

$$B_\eta = A_\eta \cup (\cup \{f(\xi) : \xi \in A_\eta\}),$$

and we choose $\bar{\zeta}_\eta$ so that

$$B_\eta \subset \bar{\zeta}_\eta < \alpha;$$

such an element $\bar{\zeta}_\eta$ of α exists because $|A_\eta| < \alpha$ and $|f(\xi)| \leq |\rho| < \alpha$ for $\xi \in A_\eta$ and α is a regular cardinal.

We set $\bar{\zeta} = \sup_{\eta \leq \rho} \bar{\zeta}_\eta$ and we claim that $f(\bar{\zeta}) \cap A_\eta \neq \emptyset$ for $\eta \leq \rho$. Indeed if $\eta \leq \rho$ and $f(\bar{\zeta}) \cap A_\eta = \emptyset$, then for $\xi \in A_\eta$ we have $\xi \notin f(\bar{\zeta})$ and also $\bar{\zeta} \notin f(\xi)$ (since

$$f(\xi) \subset B_\eta \subset \bar{\zeta}_\eta \leq \bar{\zeta};$$

thus if $\eta \leq \rho$ and $f(\bar{\zeta}) \cap A_\eta = \emptyset$, the set $A_\eta \cup \{\bar{\zeta}\}$ is free for f and a subset of $\alpha \setminus \zeta_\eta$, contrary to the maximality of A_η . The claim is proved.

For $\eta \leq \rho$ we choose $\xi(\eta) \in f(\bar{\zeta}) \cap A_\eta$. For $\eta' < \eta \leq \rho$ we have

$$\xi(\eta') \in A_{\eta'} \subset B_{\eta'} \subset \bar{\zeta}_{\eta'} \leq \zeta_\eta < \min A_\eta \leq \xi(\eta);$$

thus the function from $\rho + 1$ into $f(\bar{\zeta})$ given by $\eta \rightarrow \xi(\eta)$ is an ordered-set isomorphism. Since $f(\bar{\zeta})$ contains a set of type $\rho + 1$, we have

$$\text{type } f(\bar{\zeta}) \geq \rho + 1.$$

This contradiction completes the proof.

Notes for Chapter 1

Infinitary combinatorics and the partition calculus, which is a generalization of the Dedekind–Dirichlet pigeon-hole principle

and whose first important result is Ramsey's theorem, $\omega \rightarrow (\omega)_m^n$ for $0 < m, n < \omega$ (Ramsey, 1930, Theorem A), was introduced and developed into a whole theory by Erdős, Rado, Hajnal, and others. Comprehensive treatments are given by Erdős & Rado (1956), Erdős, Hajnal & Rado (1965), Williams (1977), and in the treatise of Erdős, Hajnal, Máté & Rado (1982).

Argyros' ramification lemma (Lemma 1.1) is from the doctoral dissertation of Argyros (1977b).

The statement that if $f \in (\omega^+)^{\omega^+}$ and $f(\zeta) < \zeta$ for $0 < \zeta < \omega^+$ then f is constant on a cofinal subset of ω^+ , is due to Alexandroff & Urysohn (1929). Generalizations were given by Dushnik (1931), Erdős (1950) (final paragraph), Novák (1950, Theorem 1) and Bachmann (1955, section 9). Neumer (1951) proved that if ξ is an uncountable, regular cardinal, if S is a subset of ξ which meets every unbounded subset of ξ closed in the order topology of ξ , and if $f: S \rightarrow \xi$ satisfies $f(\zeta) < \zeta$ for $\zeta \neq 0$, then there are cofinal $T \subset S$ and $\bar{\zeta} < \xi$ such that $f[T] \subset \bar{\zeta}$. (Such sets S are the *stationary sets* – les ensembles stationnaires – of Bloch (1953), who showed that every stationary subset of ω^+ is the union of two disjoint stationary sets.) Fodor (1955) showed that in the statement of Neumer's theorem it is enough to assume that ξ is an ordinal number with $\text{cf}(\xi) > \omega$; and (Fodor, 1956) the set T may itself be chosen stationary.

Pairs of cardinals $\langle \kappa, \alpha \rangle$ with α strongly κ -inaccessible were introduced by Fuhrken (1965) (with different terminology) in connection with a generalization of Łoś' ultraproduct theorem (cf. Comfort & Negrepointis, 1974, section 11).

The remarkable theorem of Shanin (1946a) implies the case $\kappa = \omega, \alpha$ arbitrary (regular) of Theorem 1.4 (on ' Δ -systems', or quasi-disjoint families); see also Marczewski (1947) and Bockstein (1948) for implicit statements, and Mazur (1952, vii, p. 235) for an explicit statement and proof, of the case $\kappa = \omega, \alpha = \omega^+$ of Theorem 1.4 and for applications to product spaces.

Erdős & Rado (1960) prove Theorem 1.4 for non-limit cardinals α . Short proofs of this case were given also by Michael (1962) and Davies (1967). Theorem 1.4 in its present general form is from Erdős & Rado (1969). Alternative proofs include an unpublished

proof of Davies, referred to by Erdős & Rado (1969), and the proof of Comfort & Negrepontis (1972a, Theorem 2.1); this latter is related to arguments given by Całczyńska-Karłowicz (1964), Marek (1964), and Mostowski (1969, Theorem 13.3.1). These are given in detail, together with the converse of the theorem, due also to Erdős & Rado (1969), by Comfort & Negrepontis (1974, section 3).

The first proof given here of Theorem 1.4 is from the doctoral dissertation of Argyros (1977b); the second proof is derived from an argument of Hajnal & Máté (1975, Theorem 2.3).

Preliminary formulations of the arrow relation of Theorem 1.5 appear (with different hypotheses) in Erdős & Rado (1956, Theorem 39 (ii)) and Kurepa (1959a). A different proof of Theorem 1.5, using the saturated structures of model theory, has been announced by Simpson (1970) and appears in Keisler (1971, section 14). We note that the case $(2^\alpha)^+ \rightarrow (\alpha^+)_\alpha^2$ appears in Erdős (1942, Theorem I). Related results on partition calculus were also proved independently by Kurepa (1939, 1959a,b) (cf. the review by Erdős, 1962).

The first proof of Theorem 1.5 given here is due to the authors, while the second was shown to us by Hajnal.

Theorem 1.7 is due to Erdős & Rado (1956, Corollary 1). The present proof, based on Argyros' lemma (1.1), is due to the authors; clearly another proof is available, along the lines of the second proof of Theorem 1.5, using Lemma 1.3 (the pressing-down lemma).

The argument used in Lemma 1.8 is essentially the same as the one given by Noble & Ulmer (1972, Lemma 1.1 (ii)).

The extension of the Erdős–Rado theorem on quasi-disjoint families to singular cardinals (Theorem 1.9), already noted in preliminary form in the original paper of Shanin (1946a), is stated in Shelah (1977, Claim 2.4(B)) for $\kappa = \omega$, and was proved independently in its present general form by Argyros in his doctoral dissertation (1977b, Theorem 1.3); see also Argyros & Negrepontis (1977, 1980). Hagler and Haydon, independently, have also noted the result for $\kappa = \omega$ (letter to the authors of October, 1977).

Argyros (1977b, Theorem 1.3) also proves a converse to Theorem 1.9.

For some versions of the Erdős–Rado theorem on quasi-disjoint families which use roots of cardinals, consult Verbeek

(1970), Juhász (1971), and O'Callaghan (1975). Argyros (1980) gives a strong analogue of Theorem 1.10 with (weak) roots.

Results analogous to Theorem 1.11 have been obtained as follows: for ρ a cardinal and α regular, by Lázár (1936) and Piccard (1937); for ρ a cardinal and α a (regular or singular) cardinal, assuming the generalized continuum hypothesis, by Erdős (1950); for ρ and α cardinals, by Hajnal (1961); for ρ an ordinal and α a cardinal, by Erdős & Specker (1961). Related results and generalizations are given by Erdős & Fodor (1957) and by Erdős, Galvin & Hajnal (1975).

We note that for limit cardinals α (regular or singular), the conclusion of Theorem 1.11 is immediate from the theorem of Hajnal (1961).

For a generalization of Hajnal's theorem involving finitely additive measures, see Tsarpalias (1977, 1978a, 1980).

The combinatorial methods and theorems given in this section have extensive applications outside the scope of this monograph. In topology, for example, there are applications by Hajnal & Juhász (1967, 1969), Juhász (1971, 1976), Ginsburg & Woods (1976), Fleissner (1978) and Gerlits (1980a, b). In functional analysis there are: an example of an Eberlein compact space that cannot be embedded in a Hilbert space in its weak topology, given by Benyamini & Starbird (1976); results of Talagrand (1978, 1980) on the non-existence of strong lifting; results (described in the Notes for Chapter 5) on the embedding of $l^1(\Gamma)$ into Banach spaces $C(X)$, and in the isomorphic theory of Banach spaces; results on the conjecture of Pełczyński (1968) by Haydon (1977, 1978), Argyros & Negrepontis (1980), and Argyros (1981b); and many more.